Volt: Searching the Option Strategy Space

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Abstract

An option chain supports many more portfolios than the named templates usually used to search it. Direct construction from listed contracts makes that wider space available, but realistic evaluation is costly: each American-option portfolio may require pathwise marks, fills, transaction costs, exercise, assignment, and settlement. Volt evaluates a sample of feasible portfolios on historical or generated paths and uses the outcomes to train an order-invariant surrogate over signed contract positions. The surrogate ranks the unevaluated pool, allowing more of the contract-level search space to be considered for a given evaluation cost. Deep Sets supplies the first model. A relation-aware graph network tests whether explicit position, underlying, expiry, and account relations add value; a QUBO formulation is tested for batch selection. This paper specifies the financial assumptions, mathematical formulation, and empirical criterion: recovery of high-ranked portfolios against the cost of full path evaluation.

1 Introduction

A common way to make option-strategy search tractable is to begin with a template: choose a vertical, straddle, butterfly, calendar, or another familiar shape, then search over its strikes, expiries, and quantities. This approach reduces the number of portfolios under consideration, but the reduction occurs before their behavior is evaluated. The number of legs, option rights, and sign pattern have already been chosen by the template.

The alternative considered here starts with the listed contracts themselves. A candidate portfolio assigns a signed quantity to each eligible contract, subject to limits on exposure, liquidity, capital, and leg count. Conventional strategy names can be applied afterward for interpretation. They play no role in candidate generation, so the search includes structures outside a predefined catalogue.

The resulting search space grows quickly. With d eligible contracts and quantities from a finite set $\mathcal{Q}$, the unfiltered space contains $|\mathcal{Q}|^d$ quantity vectors. Most applications cannot evaluate every vector across every relevant market path. Evaluation is especially costly for American options because portfolio behavior depends on the path of option values and on decisions or events that occur before expiry. A terminal payoff diagram supplies the expiration geometry; path evaluation supplies the pre-expiry behavior.

Payoff diagrams remain a useful first description of the economic position and are standard in options reference material [1]. Figure 1 shows how two contract positions combine. The portfolio can be recognized as a bull call spread after it has been constructed; candidate generation uses the two signed positions.

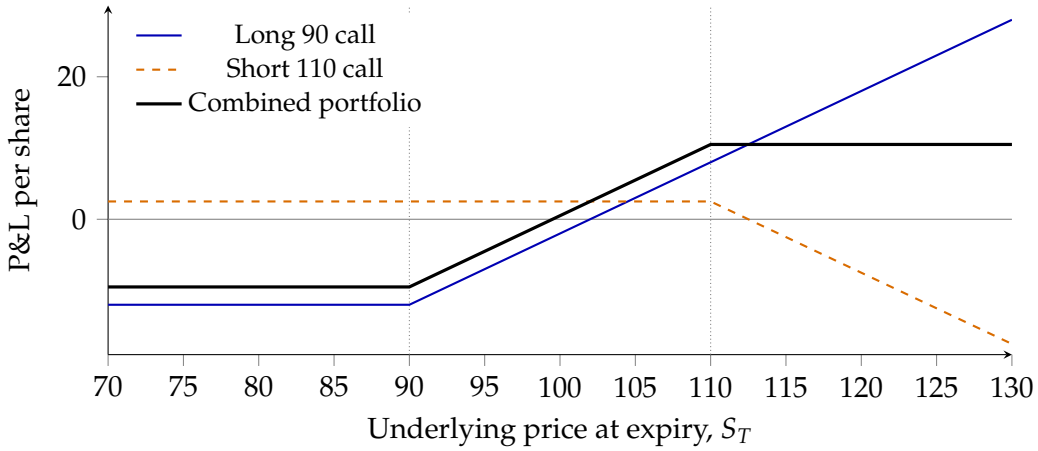


Figure 1: Expiry profit and loss for one long 90 call and one short 110 call, using illustrative entry prices of 12 and 2.50 before fees. The combined position has a net debit of 9.50, a break-even price of 99.50, a maximum loss of 9.50, and a maximum profit of 10.50 per share. The pathwise evaluator additionally requires pre-expiry marks, exercise, assignment, fills, and management actions.

The practical objective is to recover high-utility portfolios without applying the full evaluator to the entire combinatorial set. A small evaluated sample is used to fit a surrogate; the surrogate ranks the remaining pool; an acquisition rule selects another batch for the financial evaluator; and the process repeats. Success is measured against exhaustive evaluation on small problems and by held-out performance on later decision dates.

This paper develops that problem in three stages. Section 2 describes the trading and derivatives assumptions. Sections 3 and 4 give the mathematical formulation and the first learning model. Sections 5 and 6 define candidate acquisition and the empirical comparison. The initial study searches entry portfolios under a fixed management rule. Sequential management is discussed separately in Section 9.

2 Trading and derivatives setting

2.1 Portfolio construction and strategy management

An entry portfolio specifies the contracts and signed quantities held at the decision time. A management rule specifies later decisions such as closing, resizing, rolling, exercising, or waiting. Assignment and other external events can change the position independently of the holder's choices.

The first experiment searches entry portfolios while holding the management rule and external-event model fixed. This isolates the portfolio-construction question: given the same information, paths, execution assumptions, and management rule, which initial combinations warrant further evaluation? Changing the management rule defines another experiment.

2.2 A concrete decision state

Suppose the point-in-time chain shows spot $S_t = 100$, a 90 call marked at 12, and a 110 call marked at 2.50, with both calls expiring in 30 days. Spot and marks are market observations. The listed strikes, exercise terms, expiry, and contract multiplier are contract data. The portfolio decision is the signed quantity on each contract: $+1$ for the 90 call and -1 for the 110 call in Figure 1.

The marks would ordinarily come from a timestamped option chain under a declared quote rule, such as mid, executable bid or ask, or a fitted surface. Synthetic examples set these values directly. The same applies to spot, interest rates, dividends, and the initial volatility state.

Changing a strike, expiry, option right, or quantity produces another candidate portfolio. Both leg orders map to the same candidate. A new date or a new set of contemporaneous marks produces another decision state. In the graph comparison, one model input represents a candidate–state pair, so the same contract quantities observed on two dates give two examples with different market features.

Candidate portfolios are generated by software from the eligible chain and the declared quantity, leg-count, liquidity, exposure, and capital limits. Manual entry remains useful for fixtures such as Figure 1.

2.3 Where market paths come from

The evaluator accepts three kinds of path evidence:

- historical paths assembled from point-in-time underlying and option observations;
- generated paths sampled from a stated stochastic or stress model; and
- forecast-conditioned paths produced by a model fitted before the decision time.

Historical replay measures behavior on recorded observations. Generated paths measure behavior under the specified distribution or stress construction. Forecast-conditioned paths measure behavior under a forecast that was available when the portfolio was chosen.

A path of spot, volatility, rates, and dividends still requires option valuation at each observation. Historical studies may use recorded option quotes. Generated studies require a pricing method that maps the factor state and contract terms to an option mark and, when execution is modelled, bid and ask prices. The pricing method, quote-selection rule, and missing-data policy are part of the experiment.

2.4 Early exercise and assignment

Let

$$\mathcal{T} = \{\tau_0, \tau_1, \ldots, \tau_H\}, \qquad \tau_0 < \tau_1 < \cdots < \tau_H,$$

be the observation grid. Every option has an exercise-opportunity set $\mathcal{E} \subseteq \mathcal{T}$. European exercise is the singleton $\mathcal{E} = \{T\}$. Bermudan exercise uses a finite declared subset. American exercise uses every grid point between the first exercise time and expiry:

$$\mathcal{E} = \{\tau_j \in \mathcal{T} : t_{\text{first}} \le \tau_j \le T\}.$$

European and Bermudan exercise are restricted cases of the same exercise-opportunity representation. The value and realized outcome of an American option depend on the observation grid and the holder's exercise rule. The treatment of a short position also requires an assignment rule or a supplied assignment schedule. The experiment therefore records:

- the observation grid;
- the holder's exercise rule;
- the short-assignment model;
- the option-valuation method;
- the fill and transaction-cost rules; and
- settlement and liquidation conventions.

Least-squares Monte Carlo is one established approach to estimating continuation values for American options [3]. It is one optional valuation component of the search method.

3 Mathematical formulation

3.1 Contracts and portfolios

At decision time t, let

$$\mathcal{U}_t = \{o_1, \ldots, o_d\}$$

be the eligible listed contracts. Contract o_i contains its identifier, underlying, currency, option right, strike, multiplier, exercise set, expiry, and settlement rule.

A portfolio is a quantity vector

$$q = (q_1, \ldots, q_d) \in \mathcal{Q}^d, \qquad \mathcal{Q} \subset \mathbb{Z},$$

where $q_i > 0$ is long, $q_i < 0$ is short, and $q_i = 0$ omits the contract. Portfolio constraints define the candidate set

$$\mathcal{C}_t = \{q \in \mathcal{Q}^d : g_j(q, c_t) \leq 0,\ j = 1, \ldots, J\},$$

where c_t is the information available at the decision time. The constraints may include gross quantity, leg count, liquidity, capital, margin, and exposure bounds.

The vector q contains no construction order. A sequence of trades is converted to q by summing quantities for each contract. Reordering the same trades leaves q unchanged. Any portfolio model used here must therefore be invariant to the order in which its nonzero positions are serialized.

3.2 Paths and portfolio outcomes

One path is

$$\xi_m = (z_{m,0}, z_{m,1}, \ldots, z_{m,H}),$$

where $z_{m,j}$ contains the market observation, contract prices, and external events at time τ_j. Let ν denote the option-valuation or quote-selection rule, μ the management and exercise rule, α the assignment rule, and ω the fill, cost, accounting, and settlement rules.

The financial evaluator returns

$$R(q, \xi_m; c_t, \nu, \mu, \alpha, \omega) \in \mathbb{R}^p.$$

The result contains leg-level records and portfolio aggregates. Components may include net P&L, pathwise equity, drawdown, turnover, transaction costs, cash, delivered underlying, exercise effects, assignment effects, and terminal inventory.

For a cohort of M paths,

$$\Xi_t = \{\xi_1, \ldots, \xi_M\},$$

an aggregation rule A creates the model target

$$y_t(q) = A\left(\{R(q, \xi_m; c_t, \nu, \mu, \alpha, \omega)\}_{m=1}^{M}\right) \in \mathbb{R}^r.$$

The target can retain several outcome and risk measures. A stated utility function $U(y_t(q))$ supplies a ranking when the experiment requires one ordering.

Model inputs contain q, c_t, and forecast artifacts available at time t. Future observations in Ξ_t are used to compute outcomes. Training, validation, and test periods follow the decision clock, with a purge interval where label horizons overlap.

3.3 Budgeted search

Evaluating every candidate on every path costs approximately

$$|\mathcal{C}_t|\, M\, C_R,$$

where C_R is the average cost of evaluating one candidate on one path. Let B be the number of candidates that can receive full path evaluation, with $B < |\mathcal{C}_t|$. The search procedure chooses an evaluated subset $\mathcal{L}_t \subset \mathcal{C}_t$, $|\mathcal{L}_t| \le B$, and returns the best evaluated candidate according to U.

On a small candidate pool, exhaustive evaluation identifies

$$q_t^\star = \arg\max_{q \in \mathcal{C}_t} U(y_t(q)).$$

This gives a reference for simple regret,

$$U(y_t(q_t^\star)) - \max_{q \in \mathcal{L}_t} U(y_t(q)),$$

and for recovery of the top k candidates. These quantities make the cost and benefit of learned acquisition observable.

4 Permutation-invariant portfolio models

4.1 Deep Sets baseline

For each nonzero position, construct a feature vector

$$x_i = x(o_i, q_i, c_t).$$

Features can include option right, signed quantity, strike relative to spot, time to exercise boundaries and expiry, multiplier, settlement, mark, bid–ask spread, implied-volatility coordinates, and liquidity measures.

The first nonlinear model is

$$h_i = \phi_\theta(x_i, c_t),$$
$$h(q, c_t) = \sum_{i:q_i \neq 0} h_i,$$
$$\widehat{y}_t(q) = \rho_\theta(h(q, c_t), c_t).$$

The shared leg map ϕ_θ and the sum make the output invariant to position order. The signed quantity remains part of each position feature, and the sum can retain information about gross scale and number of positions.

Zaheer et al. characterize permutation-invariant set functions under their stated domain conditions and motivate this form [2]. The application here follows from the financial definition of a portfolio as a quantity assignment without a leg ordering.

The model predicts the outcome vector $y_t(q)$. Training may combine component-wise regression with a pairwise ranking loss among candidates from the same decision state. Calibration of the outcome components and quality of the induced ranking are reported separately.

4.2 Graph comparison

For a candidate q at decision state c_t, the proposed graph is

$$G(q, c_t) = (V, E), \qquad V = V_{\text{pos}} \cup V_{\text{underlying}} \cup V_{\text{expiry}} \cup \{v_{\text{account}}\}.$$

Each nonzero quantity creates one position node with feature vector x_i. Typed edges connect that position to its underlying, expiry, and account. Optional position-to-position edges record declared relations such as common expiry, common strike, or ordered strikes within one option right. Market features belong to the underlying and expiry nodes; capital and inventory features belong to the account node.

Relation-specific message passing updates the node representations, followed by an order-invariant readout over the position nodes and a prediction head for $\widehat{y}_t(q)$. Graph construction is deterministic for a given candidate–state pair. Serializing its position nodes in another order gives the same financial input. A change to a contract quantity changes the candidate; a change to the contemporaneous market data changes the state.

This graph adds relational structure to the input. Path generation and option valuation remain responsibilities of the financial evaluator, which produces the training outcomes in both the Deep Sets and graph experiments. The graph model becomes useful if the typed relations improve prediction or candidate ranking beyond the set model [4].

The graph comparison uses the same position and market information, target, time split, parameter budget, and optimization budget as the Deep Sets model. The parameter-matched nonlinear set model controls for model capacity. Edge-removal and relation-label ablations then test whether the graph relations account for any observed difference.

5 Selecting candidates for evaluation

After an initial evaluated sample, the model predicts outcomes for the remaining pool. An acquisition score can combine predicted utility, uncertainty, coverage of sparsely sampled regions,

and path-evaluation cost. The simplest rules select candidates at random, by predicted utility, by uncertainty, or by a greedy utility–diversity score. These establish the baseline for a batch optimizer.

5.1 QUBO batch selection

Suppose the frontier contains n candidates. Let $b_i \in \{0, 1\}$ indicate selection of candidate i, let a_i be its acquisition value, and let $s_{ij} \geq 0$ measure similarity between candidates i and j. For an equal-cost batch of size k, consider

$$\min_{b \in \{0,1\}^n} -\sum_i a_i b_i + \gamma \sum_{i<j} s_{ij} b_i b_j + \rho \left(\sum_i b_i - k \right)^2 .$$

The linear term favors candidates with high acquisition value. The quadratic term discourages a batch concentrated in one region of the portfolio space. The final term encodes the batch size.

QUBO is a binary optimization formulation with classical and quantum-inspired solution methods [5]. Its role here is candidate allocation. The financial evaluator supplies outcomes and the constraint system supplies admissibility. The empirical comparison charges matrix construction and solver time to the acquisition method and includes exhaustive solutions on small frontiers, classical heuristics, greedy diversity, and random selection.

5.2 Learning loop

For each decision state, the search proceeds as follows:

1. generate and constrain the candidate pool;
2. evaluate an initial sample over the path cohort;
3. fit the portfolio model;
4. score the remaining candidates;
5. select another batch using one acquisition rule;
6. evaluate the selected batch and update the training data; and
7. stop at the declared evaluation or wall-clock budget.

Every round records the candidate pool, evaluated subset, model checkpoint, scores, selection rule, path cohort, financial assumptions, and results.

6 Empirical design

6.1 Initial study

The initial study uses listed American-style options on one underlying. The candidate generator selects contracts and quantities directly, subject to declared bounds on moneyness or delta, expiry, quantity, spread, volume, open interest, and leg count. Analysis may add strategy labels after generation.

The study fixes one entry time and management rule for every candidate at a decision date. The exercise rule, assignment process, observation grid, valuation method, and transaction-cost model

are shared across candidates. Historical and generated path cohorts are analysed separately because they support different interpretations.

A small universe is evaluated exhaustively. It provides the target ranking and shows where portfolio symmetry and offsetting positions reduce duplicate constructions. Later experiments expand the number of contracts, legs, paths, and decision dates one dimension at a time.

6.2 Comparisons

Prediction models include linear and tree models over portfolio aggregates, the Deep Sets model, and a parameter-matched graph model. Acquisition methods include random sampling, stratified sampling, predicted utility, uncertainty, greedy diversity, and QUBO. Every method receives the same candidate pool, initial evaluated sample, path cohort, and compute accounting.

Primary measurements are top-k recovery, simple regret, rank correlation, outcome calibration, path-evaluator calls, total paths evaluated, and elapsed time. Secondary measurements include coverage across exercise and assignment states, diversity of the evaluated portfolios, transaction-cost sensitivity, and stability under alternative exercise and assignment rules.

6.3 Time and reproducibility

Decision dates are divided chronologically into training, validation, and test periods. Overlapping outcome horizons are assigned to one split or separated by a purge interval. Feature scaling and model selection use the training and validation periods. The final configuration is applied once to the test period.

Each result records the listed universe, decision timestamp, candidate constraints, path source, pricing and quote rules, exercise and assignment rules, execution assumptions, target definition, model version, random seeds, and selection history. Altering any of these inputs creates another experiment record while the portfolio quantity vector remains available for comparison.

7 Pilot integration results

The pilot tests the complete identity–evaluation–learning path on one fixed seed. It uses twelve American contracts, unit quantities, at most three legs, and a bound excluding portfolios whose expiration loss diverges as spot rises. Software generates 1,335 canonical portfolios. Each quantity vector passes through the Volt contract validator, portfolio canonicalizer, payoff projection, and graph exporter. The research tensor joins implied volatility and liquidity fields to the resulting Volt portfolio hash.

The generated decision state has spot 100, 30 calendar days to expiry, a 22-point grid, and 512 GBM paths. A CRR lattice supplies American marks and holder exercise decisions; the same decisions generate short assignment. The target contains mean P&L, lower-tail mean, P&L standard deviation, and profit probability. Utility is mean P&L plus one quarter of the lower-tail mean. The model sees 20% of the exhaustive labels initially and acquires another 10%, using 205,312 candidate–path evaluations instead of 683,520.

On the remaining candidates, Deep Sets attains Spearman rank correlation 0.972, recovers all of the exhaustive top 20 after acquisition, and finds the exhaustive optimum. Ridge regression on summed leg features attains 0.386 rank correlation and recovers 18 of the top 20. The selected generated portfolio is long one 105 call and long one 110 call. It has no exact match in the deliberately small reference catalogue used by the explanation layer. That observation identifies a candidate for further comparison; it supplies no novelty or robustness result.

The first market-data check uses the SPY chain at the 14 July 2026 close and options expiring 24 July. Daily option closes are proxy marks. Six calls and six puts nearest spot pass the declared volume, transaction-count, intrinsic value, and American implied-volatility filters. A 512-path cohort uses the eight validated daily observation times and the median selected-contract implied volatility. The 22 July equity partition is absent and is recorded in the manifest rather than imputed.

Deep Sets reaches rank correlation 0.996, full top-20 recovery, and zero simple regret under the generated cohort. The corresponding ridge values are 0.189, four of 20, and 0.0447 per share. The selected position is short one 750 call and long one 753 call, an ordinary bear call spread identified after generation. SPY closes at 738.93 on expiry; under the proxy-mark, funding, exercise, assignment, and cost rules, the position earns 1.989 per share. Across all candidates, cohort utility and this one realized-path P&L have rank correlation 0.801. This run checks the pipeline; return estimation requires executable quotes and multiple decision dates.

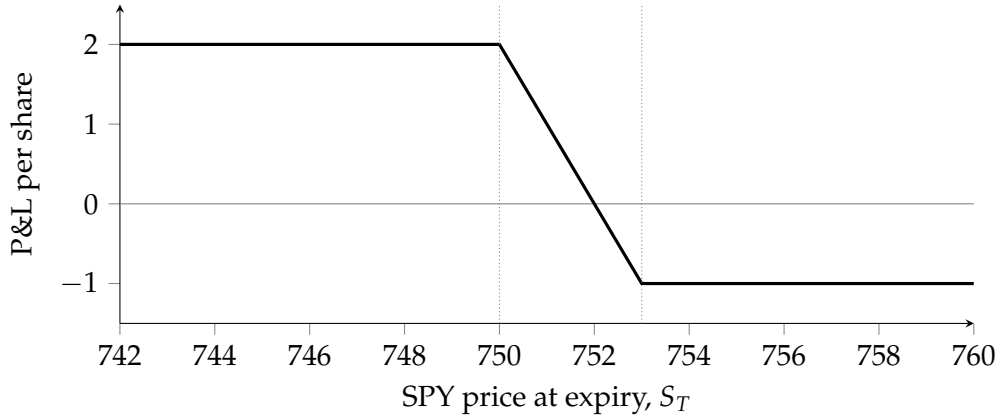


Figure 2: Expiration P&L for the SPY pilot selection: short one 750 call and long one 753 call for a 2.00 proxy credit, before financing and transaction costs. The maximum profit is 2.00, the maximum loss is 1.00, and the break-even spot is 752.

Three energy continuation checks use USO, XLE, and XOM chains over the same 14–24 July interval with lower declared liquidity thresholds. The selected positions are respectively a 119/121 bull call spread, a 56/57 bull call spread, and a 141/142 bull put spread. All three are conventional verticals despite catalogue-free generation. They earn positive P&L on the single observed path, while the cross-candidate rank correlations between generated-cohort utility and realized P&L are -0.530, -0.635, and -0.623. The surrogate accurately learns the supplied cohort labels; these negative realized correlations direct the next work toward path calibration and multiple decision dates.

Two earlier decision cohorts are registered for that multi-date test. The tariff cohort fixes information at the 1 April 2025 close and observes the 2–9 April response window described in the corresponding market study [6]. The initial US–Iran cohort fixes information at the 27 February 2026 close and

begins its response window on 2 March [7]. These dates define the information sets and evaluation windows; causal attribution requires a separate design. The tariff archive currently lacks the completion certificates required for a primary result. The Iran option aggregates are recoverable from source files, but their derived partitions still require reconciliation.

Table 1: Fixed-seed pilot ranking diagnostics. Surrogate correlations compare predicted and exhaustive generated-cohort utility on candidates outside the initial evaluated set. Realized correlation compares exhaustive cohort utility with P&L on one later observed path.

Cohort	Deep Sets ρ	Ridge ρ	Top-20 recall	Realized ρ
Generated	0.972	0.386	1.00	—
SPY	0.996	0.189	1.00	0.801
USO	0.996	0.221	1.00	-0.530
XLE	0.968	0.231	1.00	-0.635
XOM	0.967	0.277	1.00	-0.623

Validated NYMEX session data supplies contemporaneous CL, RB, and BZ evidence for the July energy checks. This evidence is constant across candidates in one decision state, so it is recorded for explanation and future multi-date conditioning rather than credited with the within-state ranking. The available research inventory contains outright futures and no identified options-on-futures series. Futures options would require contract, exercise, delivery, margin, and settlement semantics beyond the current equity-option pilot.

Every selected portfolio produces a versioned evidence packet containing its exact positions, mark authority, payoff geometry, entry Greeks, path statistics, leg attribution, exercise and assignment counts, and limited catalogue comparison. Human-facing text is rendered from those computed fields. An LLM explanation is required to bind every numerical claim to the packet and to reserve claims about execution, causal event attribution, robustness, and innovation for the corresponding evidence.

8 Current implementation

Volt currently supplies the financial identity and accounting components needed by this formulation. Contracts carry European, Bermudan, or American exercise schedules. Position netting converts repeated positions into a single quantity per contract. The runtime handles close, exercise, assignment, expiry, cash, delivered underlying, and journaled state changes. The market-data layer gives historical and generated option observations a shared versioned interface with declared price authority. Scenario paths retain ordered points, source identity, and reproducibility seeds. The scenario evaluator consumes supplied option marks or quotes and reports fills, costs, pathwise P&L, terminal P&L, and leg attribution. Candidate actions resolve strike, moneyness, delta, and expiry buckets to listed contracts.

The supervised graph example evaluates a bounded terminal-payoff fixture. A general graph-export command now accepts declared marked contracts and quantity vectors, then returns Volt portfolio hashes and graph samples. The pilot research driver adds a generated cohort builder, CRR American marks, fixed exercise and assignment policies, constrained candidate enumeration,

path-target aggregation, Deep Sets and ridge baselines, one batch-acquisition step, market-data adapters, and evidence packets.

The next integration work is:

1. reconcile and certify the older event archives, then add several chronologically separated decision dates;
2. add fixed-horizon closing and scheduled or banded delta hedging for gamma attribution;
3. replace close proxies with executable quote rules where quote coverage permits;
4. run iterative random, uncertainty, utility, and diversity acquisition baselines;
5. compare a parameter-matched relation-aware graph model with Deep Sets; and
6. test QUBO batch selection after the classical acquisition baselines.

9 Extension to sequential control

Static discovery chooses an entry portfolio under a fixed management rule. Sequential control chooses actions as market observations, fills, exercise, assignment, and inventory change the state. Its data consist of state–action–reward trajectories.

The Volt state/action graph and legal-action mask provide part of that environment boundary. A reinforcement-learning study additionally requires a reward definition, transition scheduler, trajectory store, policy and value models, and out-of-time policy evaluation. The set or graph encoder developed for static portfolios may initialize the policy representation, while control performance determines the value of that transfer.

QUBO has a separate possible role in control: allocating a limited rollout or replay budget among frontier states. That experiment follows the static candidate-allocation comparison and uses its own baselines.

10 Discussion

The candidate pool sets the discovery domain. The path cohort, valuation method, management rule, and outcome measure set the preference represented by the labels. Reported results are conditional on these choices.

American options make this dependence visible. Early exercise changes the cash and inventory path; short assignment can create states the holder did not choose; coarse observation grids change the opportunities available to the management rule. Reported results therefore require sensitivity to the exercise policy, assignment process, grid, quotes, and execution costs.

The empirical test compares the portfolios found by a learned invariant model with those found by simple search rules at the same evaluation budget. Graph and QUBO components are measured by incremental improvement over their set-model and acquisition baselines.

11 Conclusion

Searching directly over listed option contracts expands the candidate set beyond predefined strategy templates and makes full path evaluation expensive. Volt frames the resulting task as budgeted portfolio search. A financial evaluator supplies leg and portfolio outcomes under explicit American-option assumptions; a permutation-invariant model approximates those outcomes; and an acquisition rule decides which candidates receive the next evaluations.

The first experiment compares Deep Sets with conventional portfolio models and measures ranking quality per evaluator call. Graph structure and QUBO batch selection enter as controlled extensions. Sequential management remains a subsequent state-and-action problem.

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